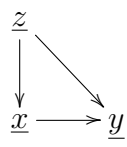


0.1 Backdoor

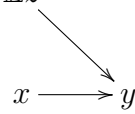
Claim 1 If $OP =$



then

$$P(y|\mathcal{D}\underline{x} = x) = \sum_z P(y|x, z)P(z) \quad (1)$$

$$= \mathbb{E}z \quad (2)$$



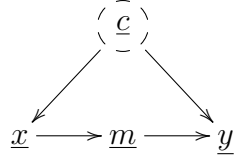
proof:

See https://github.com/rrtucci/dag_iden_detector for computer generated numerical proof.

QED

0.2 Frontdoor

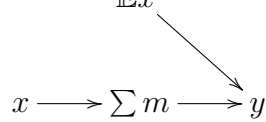
Claim 2 If $OP =$



then

$$P(y|\mathcal{D}\underline{x} = x) = \sum_m \left[\sum_{x'} P(y|m, x')P(x') \right] P(m|x) \quad (3)$$

$$= \mathbb{E}x' \quad (4)$$

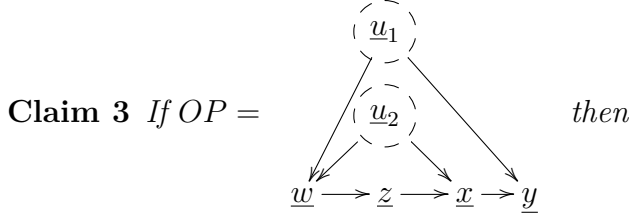


proof:

See https://github.com/rrtucci/dag_iden_detector for computer generated numerical proof.

QED

0.3 Napkin1



$$P(y \mid \mathcal{D}\underline{x} = x) = \frac{\sum_{w,z} P(x, y \mid w, z) P(w) P(z)}{\sum_y \text{num}} \quad (5)$$

$$= \frac{\sum_{w,z} P(y \mid x, w, z) P(x \mid w, z) P(z) P(w)}{\sum_y \text{num}} \quad (6)$$

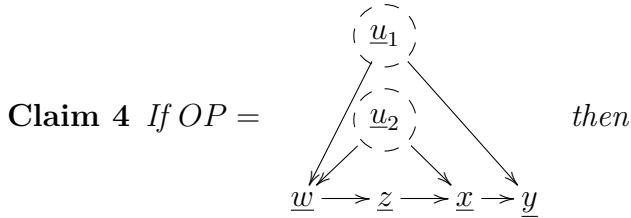
$$= \frac{1}{\sum_y \text{num}} \mathbb{E}w \begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathbb{E}z \xrightarrow{\quad} x \xrightarrow{\quad} y \quad (7)$$

proof:

See https://github.com/rrtucci/dag_iden_detector for computer generated numerical proof.

QED

0.4 Napkin2



$$P(y \mid \mathcal{D}\underline{x} = x) = \sum_z P(y \mid x, z) P(z) \quad (8)$$

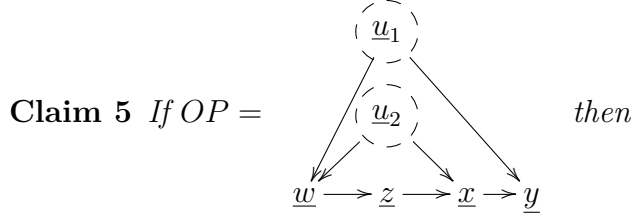
$$= \begin{array}{c} \mathbb{E}z \\ \searrow \\ x \longrightarrow y \end{array} \quad (9)$$

proof:

This claim is **INCORRECT**. See https://github.com/rrtucci/dag_iden_detector for computer generated numerical counterexample.

QED

0.5 Napkin3



$$P(y \mid \mathcal{D}\underline{x} = x) = \sum_w P(y|x, w)P(w) \quad (10)$$

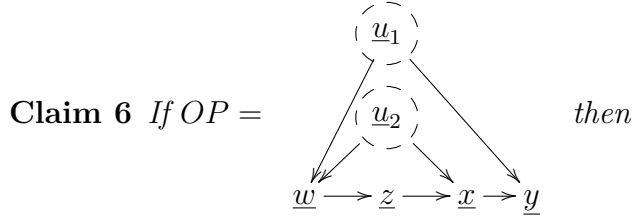
$$= \begin{array}{c} \mathbb{E}w \\ \searrow \\ x \longrightarrow y \end{array} \quad (11)$$

proof:

This claim is **INCORRECT**. See https://github.com/rrtucci/dag_iden_detector for computer generated numerical counterexample.

QED

0.6 Napkin4



$$P(y \mid \mathcal{D}\underline{x} = x) = \frac{\sum_w P(x, y|w, z)P(w)}{\sum_y num} \quad (12)$$

$$= \frac{\sum_w P(y|x, w, z)P(x|w, z)P(w)}{\sum_y num} \quad (13)$$

$$= \frac{1}{\sum_y num} \mathbb{E}w \begin{array}{c} \xrightarrow{\quad} z \xrightarrow{\quad} x \xrightarrow{\quad} y \\ \xrightarrow{\quad} x \end{array} \quad (14)$$

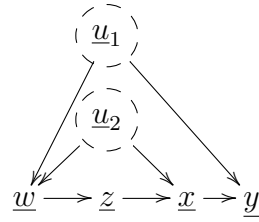
proof:

This claim is **INCORRECT**. See https://github.com/rrtucci/dag_iden_detector for computer generated numerical counterexample.

QED

0.7 Napkin5

Claim 7 If $OP =$



then

$$P(y \mid \mathcal{D}\underline{x} = x) = \frac{\sum_{w_1, w_2} P(y|x, w_1)P(x|w_2)P(w_1)P(w_2)}{\sum_y \text{ num}} \quad (15)$$

$$= \frac{1}{\sum_y \text{ num}} \quad \begin{array}{c} \mathbb{E}w_1 \\ \mathbb{E}w_2 \end{array} \begin{array}{c} \searrow \\ \searrow \end{array} \underline{x} \rightarrow \underline{y} \quad (16)$$

proof:

This claim is **INCORRECT**. See <https://github.com/rrtucci/not-do-calculus> for computer generated numerical counterexample.

QED